NETWORK TROUBLESHOOTING, MANAGEMENT, MONITORING, AND OPTIMIZATION

Task 1: Check Network Configuration Using ipconfig /all

Aim: To view the complete network configuration of my computer.

Steps:

1. Press Windows + R, type cmd, and press Enter.
2. Type ipconfig /all and press Enter.
3. Find the active Wi-Fi or Ethernet adapter. Copy the IPv4 address, subnet mask, default gateway, and DNS server addresses exactly as shown.

Task 2: Compare ping Results Before and After Disconnecting

Aim: To understand how a network disconnection affects connectivity.

Steps:

4. With the internet connected, open Command Prompt and run: ping www.google.com
5. Record the output or take a screenshot.
6. Disconnect Wi-Fi or unplug the Ethernet cable. Run the same ping command again and record the result.
7. Reconnect the network and repeat the command. Record the result after reconnection.

Task 3: Monitor Network Usage While Streaming YouTube

Aim: To observe network usage for five minutes and identify the process using the most bandwidth.

Steps (Windows):

8. Open YouTube in your browser and start a video.
9. Press Ctrl + Shift + Esc to open Task Manager.
10. Open the Processes tab and check the Network column. If needed, open Performance > Wi-Fi or Ethernet to view network activity.
11. Observe the values for five minutes. Note the highest value you observe and the process showing the most network activity.
12. Remember that the Processes tab's Network column may show a changing rate; record the units shown by Windows (for example, Mbps).

Task 4: Capture and Filter Traffic in Wireshark

Aim: To capture at least 30 seconds of traffic while browsing a website and inspect HTTP/HTTPS packets.

Steps:

13. Install Wireshark from https://www.wireshark.org/ if it is not already installed.
14. Open Wireshark and start capturing on the active Wi-Fi or Ethernet interface.
15. Browse Flipkart or Instagram Web for at least 30 seconds, then stop the capture.
16. In the display filter bar, enter: http and press Enter. Then try: tls or https if supported by the Wireshark version and capture. HTTPS commonly appears as TLS traffic; a display filter of tls is useful for encrypted web traffic.

17. Inspect the packet list and packet details. Do not expect to see encrypted page contents, passwords, or private messages in HTTPS packets.

Three types of information visible in filtered packets:

- Source and destination IP addresses, showing which endpoints communicate.
- Packet timestamps, frame lengths, and protocol information.
- For HTTPS/TLS, details such as the TLS handshake and sometimes the server name (depending on the capture and settings); the actual encrypted page content is not normally readable.

Screenshot: [Paste Wireshark screenshot showing your display filter and packets here]

Important: In current Wireshark versions, HTTPS is usually represented as TLS rather than a separate 'https' protocol label, so try the display filter tls. HTTP traffic may not appear when the site uses HTTPS.

Task 5: AI Troubleshooting Checklist for Slow Online Gaming

Prompt used: “Suggest a checklist to improve home network speed and reliability for online gaming.”

AI response:

18. Use an Ethernet cable instead of Wi-Fi when possible to reduce wireless interference and latency.
19. If using Wi-Fi, connect to the 5 GHz or 6 GHz band when supported and stay near the router.
20. Pause large downloads, cloud backups, and video streams on other devices while gaming.
21. Restart the router and modem if the connection is unstable, and check for firmware updates from the manufacturer.
22. Place the router in an open, central location away from thick walls and sources of interference.
23. Check for packet loss and latency using ping; compare results at different times.
24. If many devices share the connection, use Quality of Service (QoS) settings if the router supports them.
25. If the issue continues, contact the internet provider and share speed, latency, and packet-loss test results.

Suggestion I would try first: Use an Ethernet cable for gaming, because a wired connection is generally more stable and less affected by Wi-Fi interference.

Note: I would test one change at a time and compare latency before and after the change.

Final Conclusion

In this assignment, I learned how to check network settings, test connectivity with ping, monitor network usage, capture and filter packets in Wireshark, and prepare a checklist to improve gaming network performance. I will add my actual command outputs, observed measurements, and screenshots before submitting the assignment.